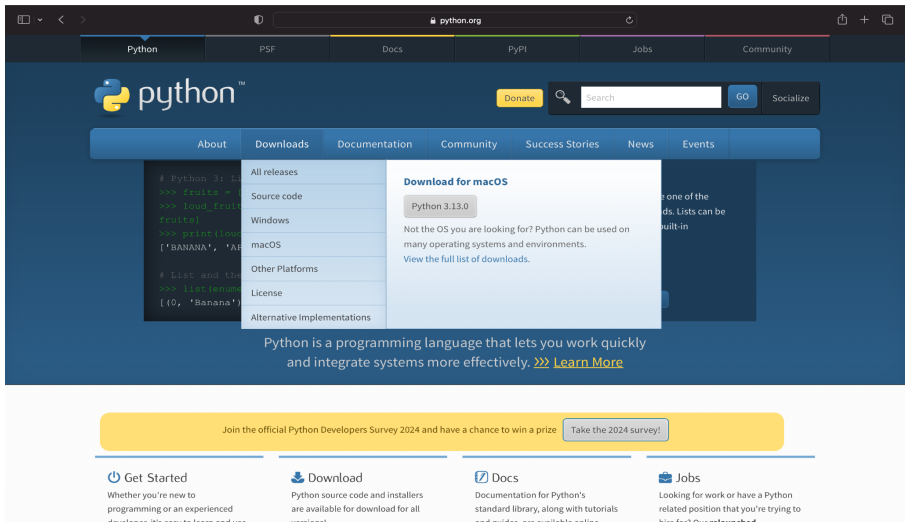
Python Basics – Lab #1 of 3

This is the first lab in a three-part series centered on the fundamentals of Python. We will explore the differences between the interactive interpreter and the IDLE environment, experiment with variables, as well as learn to create a program.

Downloading Python

1. Our first step is to go to www.python.org.
2. Next, click on the downloads section in the image below, you'll see options to download Python for MacOS, Windows, and Linux. Keep in mind that MacOS and Linux typically have Python pre-installed. Choose your operating system and download the latest version available for it.



3. On your Mac or Linux machine, you can access the pre-installed Python programming environment or interactive interpreter by opening your terminal and typing "python," then pressing enter, as shown in the example below.

```
[kwtrain ~%python

WARNING: Python 2.7 is not recommended.
This version is included in macOS for compatibility with legacy software.
Future versions of macOS will not include Python 2.7.
Instead, it is recommended that you transition to using 'python3' from within Terminal.

Python 2.7.16 (default, Aug 29 2022, 10:34:32)
[GCC Apple LLVM 12.0.5 (clang-1205.0.19.59.6) [+internal-os, ptrauth-isa=deploy on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

4. You'll see that the version in my command prompt is 2.7, which is older than the 3.13 version I downloaded. To exit Python in the command prompt, type `exit()` and then

Copyright 2024, Kevin Wallace Training

enter `python3.13` to launch the latest version of Python (at the time this documentation was created)..

```
kwtrain ~%python

WARNING: Python 2.7 is not recommended.
This version is included in macOS for compatibility with legacy software.
Future versions of macOS will not include Python 2.7.
Instead, it is recommended that you transition to using 'python3' from within Terminal.

Python 2.7.16 (default, Aug 29 2022, 10:34:32)
[GCC Apple LLVM 12.0.5 (clang-1205.0.19.59.6) [+internal-os, ptrauth-isa=deploy on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()
kwtrain ~%python3.13
Python 3.13.0 (v3.13.0:60403a5409f, Oct  7 2024, 00:37:40) [Clang 15.0.0 (clang-1500.3.9.4)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

Entering Commands using the Interactive Interpreter

In this section, we will learn how to use the interactive interpreter, which is indicated by three greater-than signs (>>>). When we enter a command in the command prompt, it is interpreted immediately.

1. To start, let's do a simple mathematical equation. In your command prompt, type in 4*4 and press Enter and let's see the results.

```
Type "help", "copyright", "credits" or "license" for more information.
>>> 4*4
16
>>>
```

2. Next, let's put 100-20 and hit Enter to see our results.

```
>>> 100-20
80
>>>
```

3. Finally, let's use mixed case (a.k.a. camel case) when naming our variables. The first line is 'firstName="Your First Name"', hit enter. The second line is 'space=" "', hit enter. The third line is 'lastName="Your last name"', then hit enter. For the final step, let's use these variables to display our name. Type 'print(firstName + space + lastName)' and then press enter.

```
>>> firstName="Kevin"
>>> space=" "
>>> lastName="Wallace"
>>> print(firstName+space+lastName)
Kevin Wallace
>>>
```

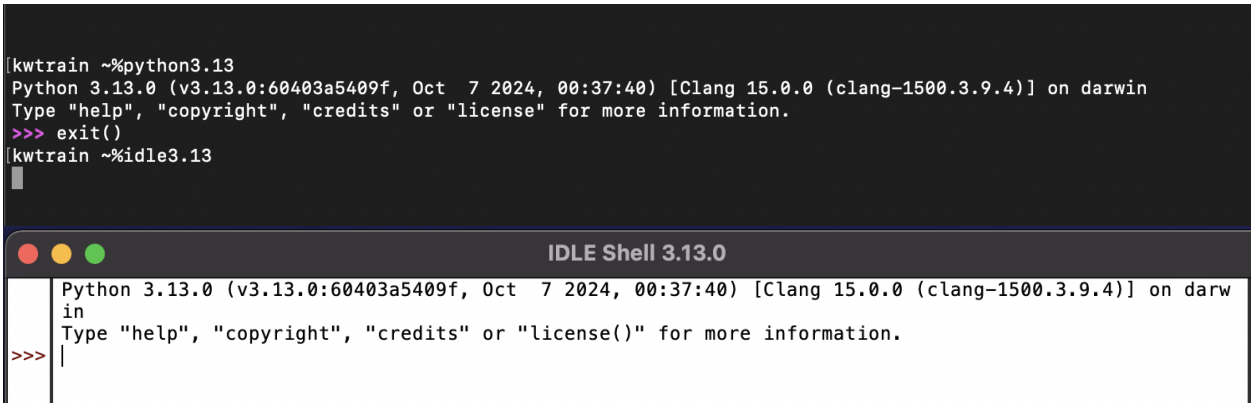
(##Please note that our first name, space, and last name are all represented as strings, which consist of a sequence of characters enclosed in quotation marks. We will explore additional data types in our second Python lab..)

Copyright 2024, Kevin Wallace Training

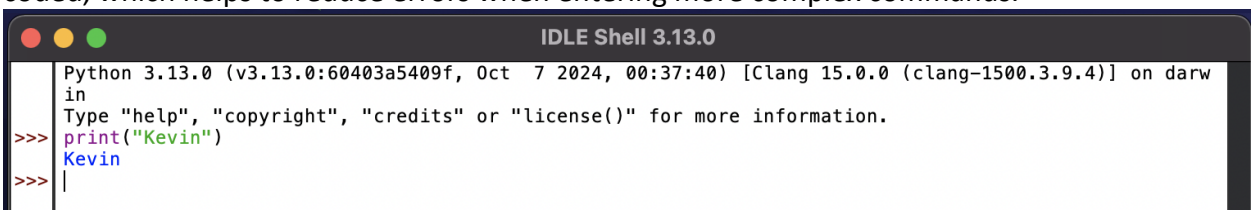
How to enter and use the IDLE environment

IDLE stands for Integrated Development and Learning Environment. While the interactive interpreter is useful for entering commands that are immediately interpreted, in real-world execution and programming, we will need to write programs that can be executed within the IDLE environment.

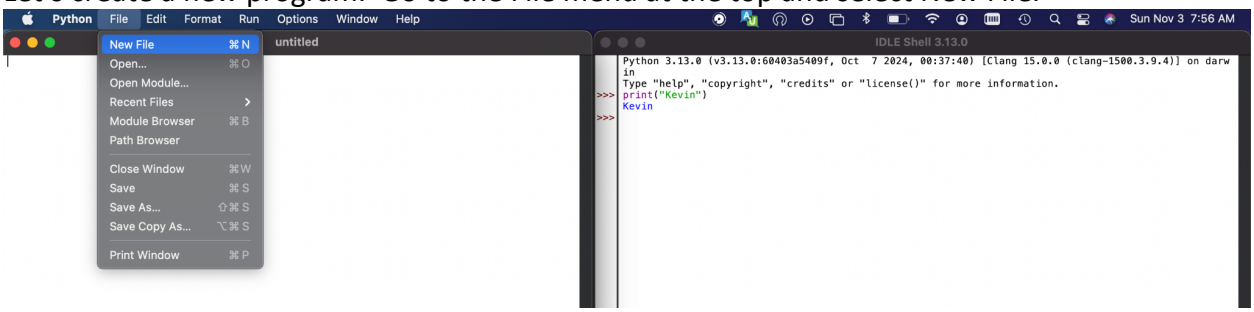
1. First, to leave the interactive interpreter and access the IDLE environment, type `exit()` and press enter. Then, enter `idle3.13` and hit enter again. Now remember, we want to run the most up to date version of IDLE, which happens to be 3.13 at the time this document was created.



2. We are now in the IDLE environment, which looks similar to the interactive interpreter. If you type `print("Your Name")` and press enter, you'll notice that the text is color-coded, which helps to reduce errors when entering more complex commands.

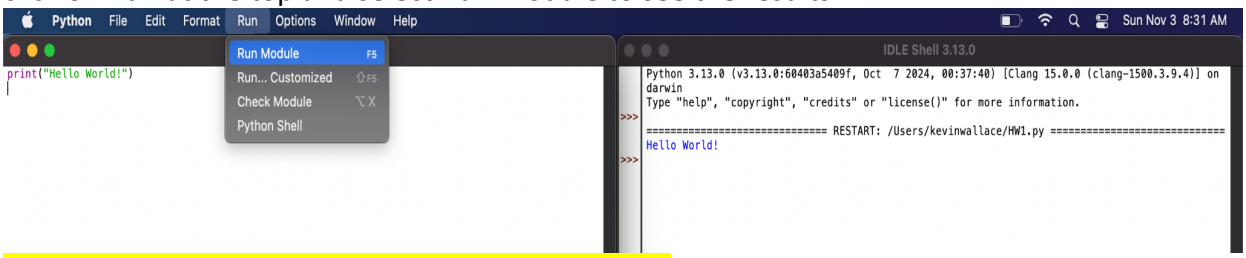


3. Let's create a new program. Go to the File menu at the top and select New File.



Copyright 2024, Kevin Wallace Training

4. In this new file, type `print("Hello World!")`. Next, save the file as 'HW1'. After that, click on Run at the top and select Run Module to see the results.



(##We have now written our first Python program!)

We will continue our learning in part 2 of this Python series…

Copyright 2024, Kevin Wallace Training